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Long-Term Adaptation among Adolescent and Young Adult Children to Familial Cancer Risk

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Abstract

Background: It is important to examine adolescent and young adult (AYA) children's long-term psychosocial and behavioral adaptation to disclosure of maternal *BRCA* positive carrier status (*BRCA*+) to inform approaches for familial cancer risk communication, education, and counseling.

Methods: Mothers underwent *BRCA* genetic testing 1–5 years earlier. Group differences in AYAs' self-reported outcomes were analyzed by maternal health and carrier status, and child age and gender.

Results: A total of N = 272 AYAs were enrolled: 76.1% of their mothers were breast/ovarian cancer survivors and 17.3% were *BRCA*+. AYAs' cancer risk behavior (tobacco and alcohol use, physical activity) and psychologic distress levels did not vary by maternal status. In bivariate analyses, AYAs of cancer-surviving mothers believed themselves to be at greater risk for, and

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Author Contributions

Dr. McDonnell drafted the initial manuscript and carried out the initial analyses.

Prof. Peshkin conceptualized and designed the study and reviewed and revised the manuscript.

Ms. DeMarco and Ms, Schneider coordinated and supervised data collection and critically reviewed the manuscript for important intellectual content.

Drs. Peterson, Arun, Miesfeldt, O'Neill, Garber, Isaacs, and Luta conceptualized and designed the study and critically reviewed the manuscript for important intellectual content.

Dr. Tercyak conceptualized and designed the study, designed the data collection instruments, drafted the initial manuscript, and reviewed and revised the manuscript.

All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

Conflict of Interest

None of the authors has any conflicts of interest to report.

Ethics Declaration

This study was approved by the Georgetown University Institutional Review Board. Informed consent and assent were obtained from all participants. All data were de-identified prior to analysis.

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were more knowledgeable about, cancer than AYAs of mothers without cancer. AYAs of *BRCA*+ mothers were more concerned about cancer, held stronger beliefs about genetic risk, and placed a higher value on learning about genetics. In adjusted models, maternal cancer history (not *BRCA*+) remained associated with AYAs' greater perceptions of cancer risk (p = .002), and knowledge about cancer (p = .03) and its causes (p = .002).

Conclusion: Disclosing maternal *BRCA*+ status did not influence children's lifestyle behavior or adversely impact quality of life long-term. AYAs of *BRCA*+ mothers were more aware of and interested in genetic risk information. Such families may benefit from support to promote open communication about genetic testing choices.

Keywords

adolescents; young adults; mothers; breast cancer risk; genetic counseling; psychosocial adaptation

Introduction

A positive *BRCA1*/*BRCA2* (*BRCA*+) germline genetic test (i.e., pathogenic variant) in one member of a kindred is often a trigger for cascade testing, especially among younger and unaffected at-risk relatives.[1] Adolescent and young adult (AYA) At-risk relatives may be concerned about the possibility that they, themselves, could develop breast/ovarian or other cancers.[2] Ongoing support and attentiveness to AYA at-risk relatives is important given the rapid biopsychosocial development taking place during this phase of life.[3] While some prior research has shown that early knowledge of familial cancer risk may foster improved health behavior during adolescence,[2,4] adolescent females from high risk families could perseverate about their breast/ovarian cancer risk and perceive greater hazards to themselves than do their male counterparts.[2,5]

The information and support needs of AYA at-risk relatives of mothers with breast cancer portends that the psychosocial impact of diagnosis and treatment on children can be significant.[6,7] For children raised in families where *BRCA* testing is pursued, their early life experiences are often influenced by significant family histories of cancer, even in the absence of a maternal cancer diagnosis.[4,5,8] Thus, it is critical to examine the psychosocial adaptation of AYA at-risk relatives under these circumstances, especially in light of the propensity for mothers who are genetically tested to disclose carrier status information to their offspring.[9,10] Family communication patterns about cancer genetics are complex, as it can be difficult for mothers to effectively share cancer risk information with younger at-risk relatives in ways that are truthful but nonthreatening.[11]

Although the presence of a *BRCA* pathogenic variant in mothers could motivate offspring to undergo cascade testing upon reaching maturity, such testing is typically not recommended during childhood given the lack of screening and risk-reducing actions at this life stage.[12,13] Despite the implications of maternal carrier status for AYA at-risk relatives, there is a very limited body of research about their short- and long-term psychosocial adaptation.[14] Some AYAs of *BRCA*+ mothers are grateful to learn the news, while others experience apprehension about its health implications for themselves. AYAs whose mothers receiving

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negative or uninformative negative results may also express relief, as well as concern over unknown causes of their families' cancer.[15]

In light of this, the aim of the present study was two-fold: (1) to characterize the long-term psychosocial and behavioral adaptation of AYA children of mothers who had previously undergone *BRCA* genetic testing; and (2) to examine possible differences in these outcomes based on AYA and maternal characteristics, including *BRCA*+ status.

Materials and Methods

Recruitment

Data were drawn from a cross-sectional, mixed-methods study involving members of English-speaking households with mother-child dyads where mothers: 1) underwent clinical genetic counseling and testing that included *BRCA* 1–5 years previously; 2) were not currently receiving treatment for cancer and/or had not a recurrence; and 3) had disclosed their *BRCA* status to their AYA children (currently 12 to 24-year-old). If a pathogenic variant was present, then the children themselves must not have yet been genetically tested to be study-eligible (i.e., absence of cascade genetic testing).

Participants were recruited from four regional cancer centers in Scarborough, ME, Boston, MA, Washington, DC, and Houston, TX. Institutional Review Board approvals were obtained from each site. Sites discovered their own cohorts by accessing and reviewing institutional electronic medical records, genetic counseling records and pedigrees, and administrative and research databases to identify potentially eligible mothers. Potentially eligible mothers were contacted by mail and telephone from local research staff explaining the purpose of the study, enumerating the household (including ages and genders of AYA offspring), screening for eligibility, and obtaining written informed consent/assent. One index child per household was chosen at random to participate (using the next birthday method[16]) to avoid selection bias. Enrolled participants then completed structured telephone interviews.

In total, N = 813 households with potentially eligible dyads were identified and approached: N = 252 (31%) could not be reached to confirm eligibility, and N = 561 (69%) had ≥ 1 successful contact. In N = 7 households the mother was deceased, and N = 168 households were determined to be ineligible. The most common reasons for ineligibility were that the mother had not already discussed cancer genetic risk information with the AYA child/ren (49%), there was no AYA in the target age range (29%), and/or the mother was undergoing active treatment for cancer or had experienced a recurrence (19%). This resulted in N=386 eligible households, and from those N = 336 (87%) agreed to participate and N = 50 declined. The primary reason for declining the study was maternal and/or AYA report(s) of not being interested in the topic (75%) or being too busy to participate (23%). Of the N = 336 participating households, N = 301 mothers (90% of those consented) and N = 274 AYAs (82% of those consented/assented) successfully completed telephone interviews. There were no meaningful site differences observed at any stage of the recruitment cascade, nor differences between the characteristics of households who consented or declined. The analytic sample was limited to N = 272 AYAs with complete data.

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Outcome Measures

Cancer risk behavior.—Tobacco and alcohol use and physical activity were measured by standard epidemiological items from the Centers for Disease Control and Prevention's Youth Risk Behavior Surveillance System.[17] Lifetime tobacco use was assessed by a yes/no item on ever having tried smoking, and lifetime alcohol use on the number of days participants had ever consumed an alcoholic beverage (0–100+). Physical activity was assessed by the number of days they had engaged in at least 60 minutes of physical activity in the past week.

Cancer cognitions.—Measures were not specific to *BRCA*-related cancers unless otherwise noted. Worries about maternal breast/ovarian cancer risk were assessed with the three-item Children's Cancer Worries Scale (CCWS).[18,19] This scale covers the extent to which children think about and/or are disturbed by their mothers' chances of developing cancer (1 = not at all or rarely, 4 = a lot; α = .75). To assess cancer risk perceptions, AYAs rated (1 = no chance, 7 = certain to happen) their beliefs about their own likelihood of developing cancer in adulthood compared to others their age.[20,21] Cancer knowledge was adapted from a previously-validated measure[22] with 22 true/false items on two subscales: 1) general cancer prevention knowledge and, 2) knowledge about the causes of cancer. Cancer prevention self-efficacy was derived using seven items on a Likert scale (1 = not at all confident, 4 = very confident; α = .76).[23] Cancer screening intentions were assessed by face-valid items on their likelihood of engaging in age- and gender-appropriate cancer screenings (1= not at all likely, 4 = very likely).[24] AYAs' perceptions about the roles of genes and health behavior in cancer development, and the importance of learning about genes and health behavior, were assessed with four task-specific items on (1 = not at all/not at all important, 7 = completely/very important).[25]

Quality of life.—Cancer-specific distress was assessed using the well-validated Impact of Events Scale (IES),[26] which measures subjective distress for a specific life event (i.e., the possibility of their mothers developing cancer [again]). The scale contains ten items (0 = not at all, 3 = often; α = .91). General life stress was assessed using the Perceived Stress Scale (PSS),[27] a 14-item measure of past 30-day stress appraisals reflecting how unpredictable, uncontrollable, and overloaded respondents find their lives (0 = never, 4 = often; α = .81). Depressive symptoms were assessed using the 20-item Center for Epidemiologic Studies Depression Scale (CES-D; 0 = none of the time, 3 = most of the time; α = .87).[28]. Anxious symptoms were assessed using the 20-item state portion of the State-Trait Anxiety Inventory (STAI; 1 = not at all, 4 = very much so; α = .90).[29]

Demographics and Clinical Covariates

Age, gender, and race/ethnicity of AYAs and their mothers were self-reported, along with maternal educational attainment and marital status. Maternal clinical characteristics, including cancer status, *BRCA* pathogenic variant status, and family proband status were reported by mothers and confirmed by chart review.

Statistical Analyses

Means and standard deviations, or frequency counts and percentages, were computed for the overall sample and compared between the 4 groups of interest, by: 1) child age

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(<18 years vs. ≥ 18 years), 2) child gender (male vs. female), 3) maternal cancer status (previously affected survivors, previously unaffected), and 4) maternal *BRCA* pathogenic variant status (*BRCA*+ vs. *BRCA*-/uninformative). Two-tailed, independent samples t-tests were used for continuous cancer cognition and quality of life outcomes, and chi-square tests of independence were conducted with cancer risk behavior outcomes and other discrete co-variates. Parsimonious multivariable regression analyses were then run for outcomes demonstrating significant associations with child and/or maternal factors.

Results

Sample Characteristics

Among the N = 272 AYAs, N = 47 (17.3%) had mothers with a known *BRCA* pathogenic variant, and N = 225 (82.7%) had mothers who either tested negative for a known familial *BRCA* pathogenic variant or received an uninformative negative *BRCA* result (Table 1). There was an approximately even split between adolescents (49.6%) and young adults (50.4%), with a mean age of 17.6 (*SD* = 3.5) years. AYAs were mostly female (67.6%) and non-Hispanic white (84.6%), with college-educated mothers (78.3%) in partnered relationships (84.9%) serving as family probands (85.2%) with a personal history of breast/ovarian cancer (76.1%; cancer survivors). Mothers with *BRCA*+ pathogenic variants were less likely to be the family proband (65.2% vs. 89.3%; $p < .001$). No other differences were observed.

Cancer Risk Behavior

Approximately one-quarter of the sample had used cigarettes in their lifetimes, more than one-third reported ever consuming at least 20 alcoholic drinks, and almost 90% did not meet public health guidelines for physical activity, which recommend at least 60 minutes of physical activity in this age group (Table 1).[30] Importantly, there were no observable differences in these outcomes when examined by maternal cancer history, maternal *BRCA* status, or child gender. Age differences in cigarette use were observed in the expected direction: young adults were more likely than adolescents to have tried smoking cigarettes, x^2 (1, N = 272) = 70.1, $p < .001$. Significant age differences were further observed for alcohol use, such that young adults were also more likely to have consumed alcohol than adolescents, x^2 (2, N = 272) = 146.7, $p < .001$.

Cancer Cognitions and Quality of Life

Data on AYAs' cancer cognitions and quality of life are given in Table 2 (overall and stratified by age and gender) and Table 3 (stratified by maternal cancer history and *BRCA* status).

Bivariate Differences in Cancer Cognitions and Quality of Life

In the overall sample, AYAs reported relatively low levels of worry about their mothers' risk of breast/ovarian cancer (*M* = 4.1 out of 12 max), although they perceived moderate risk of developing cancer themselves (*M* = 4.2 out of 7 max). Knowledge about cancer was high (*M* = 19.3 out of 24 max), with AYAs knowing about both cancer prevention (*M* = 8.0 out of 9 max) and cancer's causes (*M* = 11.3 out of 15 max). AYAs felt confident in their ability

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to take steps to prevent cancer in general (M = 22.6 out of 28 max), and endorsed strong intentions to undergo age- and gender-appropriate cancer screenings (M = 3.5 out of 4 max). AYAs thought that genes were moderately important in determining cancer risk (M = 4.9 out of 7 max), and similarly to learn about genetic risk factors for developing cancer (M = 5.4 out of 7 max). They also reported health behavior as moderately important in determining cancer risk (M = 5.1 out of 7 max), and of learning about health behavior (M = 5.8 out of 7 max). As a group, AYAs exhibited low cancer-specific distress (M = 8.2 out of 45 max), moderate general life stress (M = 23.6 out of 56 max), low depressive symptoms (M = 11.5 out of 60 max), and low anxiety (31.8 out of 80 max) that were within ranges considered normal for the population.

Regarding AYA age and gender effects, young adults perceived higher risks of developing cancer than did adolescents (p = .04). Similarly, females perceived their own cancer risk as higher than males did (p < .001). Young adults also reported being more knowledgeable about cancer than did adolescents (p = .01), especially its causes (p = .002). Conversely, adolescents were more confident in their abilities to prevent cancer than was reported by young adults (p = .02), whereas young adults had greater intentions to undergo cancer screening than did adolescents (p = .002). Finally, adolescents reported more cancer-specific distress than did young adults (p = .002), and female AYAs reported more general stress than AYA males (p <.001).

There were also significant associations with maternal status variables: AYAs with *BRCA*+ mothers reported more worry about their mothers' chances of developing cancer than did those whose mothers had received negative or uninformative genetic test results (p = .03). AYAs whose mothers were cancer survivors also perceived greater personal risks of developing cancer themselves than those whose mothers had not had cancer (p = .001). Finally, AYAs of cancer-surviving mothers were more knowledgeable about cancer than were children whose mothers had not had cancer (p = .02); this finding was especially salient for understanding cancer's causes (p = .001). Relatedly, AYA children of *BRCA*+ mothers believed more strongly that genes determine cancer risk (p = .03) and that it is important to learn about genes (p = .001) relative to AYA children of mothers with negative or uninformative genetic test results. No additional differences were observed as significant.

Multivariable Models of Differences in Cancer Cognitions and Quality of Life

As shown in Table 4, a multivariable model including child age, child gender, and maternal cancer status indicated an overall significant association with AYAs' perceived risk of developing cancer (p <.001). Specifically, females and AYA children of cancer survivors perceived themselves to be at higher personal risk of cancer than did males and children of mothers who were not previously affected.

A second model including child age and maternal cancer status indicated an overall significant association with total cancer knowledge (p = .005), such that young adults and children of cancer survivors evidenced greater knowledge about cancer than did adolescents and children of mothers not previously affected. A similar finding emerged when examining AYAs' understanding of cancer's causes (p = <.001).

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Discussion

Genetic testing for pathogenic variants in *BRCA* may influence family members' psychosocial adaptation, as measured by their cancer risk behavior, beliefs about cancer, and quality of life--especially among AYAs growing-up in households where testing is pursued and with cancer-surviving parents.[3] In this sample of AYAs whose mothers were tested for *BRCA* and who were told about their mothers' *BRCA* status, including *BRCA*+ in 17.3% of cases, outcome data collected 1–5 years post-testing failed to indicate that such disclosure was significantly associated with either health-promoting or health-compromising cancer risk behavior in these children; in addition to lack of statistical significance, differences between groups were small. That is, children who had learned of their family's high genetic risk for breast/ovarian cancer were no more or less likely to act in ways that could prevent cancer (or protect their overall health) compared to those without a *BRCA* pathogenic variant. This finding mirrors that seen elsewhere, whereby breast/ovarian cancer family history information alone is insufficient to motivate healthy behavior change,[31] even without the added influence of genetic risk information.

In contrast, several co-factors were associated with these children's long-term psychosocial adaptation. First, AYAs whose mothers survived cancer, along with female AYAs, perceived themselves to be among those at the greatest risk of developing cancer; effect sizes were medium. Given women's breast cancer risks compared to men's[32] and their odds of *BRCA*-related cancers in the presence of deleterious variants,[13] such concerns seem warranted. However, greater concern over personal cancer risk in the absence of known mutations, even among children of survivors, may be inaccurate. Thus, it is important to provide AYAs in such families with clear information about the role of genes in health, both to reduce unnecessary worry[33] and to provide tailored medical management recommendations.[13] This patient education strategy can also be decoupled from an immediate offer of genetic testing and may be viewed as more preparatory to future cascade testing upon reaching adulthood.[1]

Second, young adults and AYAs whose mothers had cancer exhibited the greatest understanding about general cancer causes; effect sizes were medium and consistent with their maturity levels and prior experience with maternal cancer.[34] Interestingly, adolescents reported higher confidence in their ability to prevent cancer, whereas young adults reported stronger intentions to undergo recommended cancer screenings; effect sizes were small to medium. That adolescents felt more confident in their ability to prevent cancer relative to young adults (who reported higher intentions to engage in cancer screening and knew more about cancer's causes), may reflect cognitive maturation in thinking about cancer risk.[34]. Although these constructs were not specific to *BRCA*-related cancers, it could ultimately translate into increased awareness of *BRCA*, proactive cancer prevention behavior in the long-term, and the pursuit of cascade genetic testing in families of known mutation carriers.

Finally, and with respect to children of *BRCA*+ mothers, they were among the most concerned about the possibility of their mothers developing breast/ovarian cancer with a medium effect size, which is consistent with epidemiologic and clinical data.[35] Likewise, these same children believed most strongly in genes determining cancer risk and the importance of learning about such risk, with female adolescents exhibiting particularly

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high levels of concern, and commensurate with their stage of life.[36] All effect sizes were medium. Taken together, these findings further underscore that AYAs generally adapt well to learning of their mothers' *BRCA* genetic risk, but subgroup differences indicate a need for more individualized educational interventions.

Limitations

The predominantly non-Hispanic white sample limits generalizability to members of underrepresented racial/ethnic groups. Future studies could incorporate more diverse samples and oversample mothers with pathogenic *BRCA* variants. Additional research could also include other comparison groups (e.g., AYAs of males/fathers tested for *BRCA*, healthy controls at average cancer risk). Furthermore, all mothers underwent comprehensive pre- and post-test genetic counseling. Thus, these findings may not extend to those receiving genetic testing without such services. Finally, other associations may exist among cancer risk behavior, beliefs about cancer, and quality of life outcomes to further inform this area of research.

Conclusions and Future Directions

Younger at-risk relatives in familial cancer kindreds are an understudied group. In this novel investigation, we found that although multiple child and maternal characteristics were mildly associated with children's cancer cognitions and quality of life, cancer risk behavior and psychological distress-related outcomes were highly resilient to *BRCA*+ notification. Taken together, these data indicate that it is safe and appropriate for mothers to share their *BRCA* genetic test results with AYA children. These data further suggest that early disclosure of maternal *BRCA+* results to AYAs about hereditary cancer and familial risk could set the stage for more open and ongoing communication about genetic testing. Additionally, these conversations could inform AYAs' further interest in and use of such testing (i.e., cascade genetic testing), as well as their engagement in other risk-reducing measures. Presently, rates of *BRCA* cascade testing are relatively low, despite the issuance of guidelines recommending such testing.[37] Such needs could be met through family-based genetic education/counseling strategies, with interventions directed toward mothers who are hesitant and/or unsure of how to communicate about genetic testing and familial risk with their offspring.[38,39] Other interventions could target AYAs themselves, encouraging discussions with their high-risk parent. These efforts to more effectively increase the uptake of cancer risk assessments, cascade testing, and age-appropriate cancer screening practices among those who are eligible will require the development and evaluation of new tools and resources delivered as adjuncts to cancer genetic counseling in light of workforce availability,[40] and the nascent state of genetic education across the lifespan. Such interventions may improve AYAs long-term psychological well-being and also encourage proactive behaviors that reduce morbidity and mortality from preventable cancers.

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Article Summary

Disclosure of maternal *BRCA* cancer risk status does not adversely influence adolescent and young adult children's lifestyle behavior, or quality of life long-term.

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What's Known on This Subject

Parents commonly disclose adult-onset cancer genetic risk information with their children, with unknown long-term impact on psychosocial and behavioral functioning.

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What This Study Adds

Disclosure of maternal *BRCA* cancer risk status does not adversely influence adolescent and young adult children's lifestyle behavior, or quality of life long-term.

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Table 1.

Sample Characteristics and Adolescent/Young Adult (AYA) Cancer Risk Behavior, by Maternal *BRCA* Status

		Maternal *BRCA* Status	
	Total Sample N = 272	***BRCA*+ N = 47**	***BRCA*−/ Uninformative N = 225**
AYA Demographics			
Age, years (M [SD])	17.6 [3.5]	17.5 [3.1]	17.6 [3.5]
< 18 years	135 [49.6%]	21 [44.7%]	114 [50.7%]
≥ 18 years	137 [50.4%]	26 [55.3%]	111 [49.3%]
Gender			
Female	184 [67.6%]	29 [61.7%]	155 [68.9%]
Male	88 [32.4%]	18 [38.3%]	70 [31.1%]
Race			
Non-Hispanic white	230 [84.6%]	37 [78.7%]	193 [85.8%]
Black, Indigenous, People of Color	42 [15.4%]	10 [21.3%]	32 [14.2%]
Maternal Demographics			
Age, years (M [SD])	49.6 [5.9]	48.7 [6.5]	49.8 [5.8]
Race			
Non-Hispanic white	239 [87.9%]	41 [87.2%]	198 [88.0%]
Black, Indigenous, People of Color	33 [12.1%]	6 [12.8%]	27 [12.0%]
Education			
< College Education	59 [21.7%]	11 [23.4%]	48 [21.3%]
≥ College Education	213 [78.3%]	36 [76.6%]	177 [78.7%]
Marital Status			
Married/Living as Married	231 [84.9%]	38 [80.9%]	193 [85.8%]
Unmarried	41 [15.1%]	9 [19.1%]	32 [14.2%]
Maternal Clinical Characteristics			
Personal History of Breast/Ovarian Cancer			
Yes	207 [76.1%]	32 [68.1%]	175 [77.8%]
No	65 [23.9%]	15 [31.9%]	50 [22.2%]
AYA Cancer Risk Behavior			

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		Maternal *BRCA* Status	
	Total Sample N = 272	***BRCA*+ N = 47**	***BRCA*−/ Uninformative N = 225**
Lifetime Tobacco Use			
Yes	74 [27.2%]	11 [23.4%]	63 [28.0%]
No	198 [72.8%]	36 [76.6%]	162 [72.0%]
Lifetime Alcohol Use (N of drinks)			
0	97 [35.7%]	18 [38.3%]	79 [35.1%]
1–19	76 [27.9%]	12 [25.5%]	64 [28.4%]
20+	99 [36.4%]	17 [36.2%]	82 [36.4%]
Meets Recommended Guidelines for Physical Activity			
Yes	34 [12.5%]	9 [19.1%]	25 [11.1%]
No	238 [87.5%]	38 [80.9%]	200 [88.9%]

Data display N [%] within a column, unless otherwise indicated; M, Mean; SD, Standard Deviation.

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Table 2.

Bivariate Analysis of Adolescent/Young Adult Psychosocial Functioning, by Age and Gender

		Age				Gender			
	Overall	**<18 years**	**≥18 years**	**Cohen's *d***	***p***	**Male**	**Female**	**Cohen's *d***	***p***
Cancer Cognitions									
Cancer Worry	4.1 [1.7]	4.0 [1.6]	4.1 [1.7]	−.1	.56	3.9 [1.6]	4.1 [1.7]	−.1	.30
Perceived Cancer Risk	4.2 [1.2]	**4.0 [1.2]**	**4.3 [1.7]**	**−.3**	**.04**	**3.8 [1.1]**	**4.4 [1.2]**	**−.5**	**<.001**
Cancer Knowledge (Total)	19.3 [2.7]	**18.9 [2.7]**	**19.7 [2.7]**	**−.3**	**.01**	19.0 [2.9]	19.5 [2.6]	−.2	.17
Prevention	8.0 [1.3]	8.0 [1.2]	8.0 [1.3]	−.1	.63	7.8 [1.4]	8.1 [1.2]	−.2	.14
Causes	11.3 [2.0]	**10.9 [1.9]**	**11.7 [1.9]**	**−.4**	**.002**	11.2 [2.0]	11.4 [1.9]	−.1	.38
Cancer Prevention Self-Efficacy	22.6 [3.5]	**23.1 [3.1]**	**22.1 [3.8]**	**.3**	**.02**	22.4 [3.6]	22.7 [3.4]	−.1	.49
Screening Intentions	3.5 [0.8]	**3.4 [0.8]**	**3.7 [0.7]**	**−.4**	**.002**	3.4 [0.7]	3.6 [0.8]	−.2	.09
Genes Determine Risk	4.9 [1.1]	4.9 [1.3]	5.0 [1.0]	−.1	.36	4.9 [1.2]	5.0 [1.1]	−.02	.87
Behavior Determines Risk	5.1 [1.2]	5.2 [1.3]	5.1 [1.1]	.1	.64	5.1 [1.3]	5.1 [1.2]	−.01	.95
Important to Learn-Genes	5.4 [1.4]	5.3 [1.3]	5.4 [1.5]	−.1	.58	5.1 [1.4]	5.5 [1.4]	−.2	.10
Important to Learn-Behavior	5.8 [1.2]	5.6 [1.2]	5.9 [1.2]	−.2	.10	5.7 [1.2]	5.8 [1.2]	−.1	.30
Quality of Life									
Cancer-Specific Distress	8.2 [8.8]	**9.8 [9.3]**	**6.5 [8.0]**	**.4**	**.002**	7.1 [8.8]	8.7 [8.8]	−.1	.19
General Life Stress	23.6 [7.5]	24.4 [7.4]	22.8 [7.5]	.2	.08	**20.8 [6.8]**	**25.0 [7.4]**	**−.6**	**<.001**
Depressive Symptoms	11.5 [8.1]	12.1 [8.6]	10.9 [7.5]	.1	.21	10.3 [7.7]	12.1 [8.2]	−.3	.08
Anxious Symptoms	31.8 [8.8]	32.3 [8.8]	31.3 [8.8]	.1	.34	31.2 [9.2]	32.1 [8.6]	−.1	.46

Data are Mean [Standard Deviation]. Bold text indicates means differed significantly at $p < .05$ in two-tailed, independent samples t-tests. Cohen's *d* is a measure of effect size.

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Table 3.

Bivariate Analysis of Adolescent/Young Adult Psychosocial Functioning, by Maternal Cancer History and *BRCA* Carriage

	Breast/Ovarian Cancer History?				*BRCA* Carriage			
	Yes	**No**	**Cohen's *d***	***p***	***BRCA+***	***BRCA−*/ Uninformative**	**Cohen's *d***	***p***
Cancer Cognitions								
Cancer Worry	4.1 [1.7]	3.9 [1.7]	.1	.31	**4.6 [2.0]**	**3.9 [1.5]**	**.4**	**.03**
Perceived Cancer Risk	**4.3 [1.2]**	**3.7 [1.0]**	**.5**	**.001**	4.4 [1.2]	4.1 [1.2]	.2	.14
Cancer Knowledge (Total)	**19.5 [2.8]**	**18.6 [2.5]**	**.3**	**.02**	19.6 [2.8]	19.2 [2.7]	.1	.38
Prevention	8.0 [1.3]	8.0 [1.2]	.01	.95	8.0 [1.4]	8.0 [1.2]	.01	.97
Causes	**11.5 [1.9]**	**10.6 [1.9]**	**.5**	**.001**	11.6 [1.9]	11.2 [2.0]	.2	.23
Cancer Prevention Self-Efficacy	22.6 [3.5]	22.7 [3.3]	−.02	.88	22.8 [3.8]	22.6 [3.4]	.1	.75
Screening Intentions	3.6 [0.7]	3.4 [0.9]	.3	.09	3.6 [0.7]	3.5 [0.8]	.01	.93
Genes Determine Risk	4.9 [1.2]	5.1 [1.0]	−.1	.38	**5.3 [1.1]**	**4.9 [1.1]**	**.4**	**.03**
Behavior Determines Risk	5.2 [1.3]	5.0 [1.1]	.1	.39	5.3 [1.2]	5.1 [1.2]	.1	.41
Important to Learn-Genes	5.3 [1.5]	5.4 [1.3]	−.1	.59	**5.9 [1.1]**	**5.2 [1.5]**	**.5**	**.001**
Important to Learn-Behavior	5.8 [1.2]	5.6 [1.3]	.2	.12	6.0 [1.1]	5.7 [1.2]	.3	.07
Quality of Life								
Cancer-Specific Distress	8.5 [8.7]	7.0 [9.2]	.2	.22	9.9 [8.6]	7.8 [8.8]	.2	.14
General Life Stress	23.6 [7.7]	23.6 [6.7]	−.002	.99	23.5 [7.7]	23.6 [7.5]	−.02	.91
Depressive Symptoms	11.3 [7.7]	12.2 [9.1]	−.2	.42	13.4 [9.7]	11.1 [7.7]	.3	.14
Anxious Symptoms	31.5 [8.4]	32.8 [9.8]	−.2	.29	32.6 [8.9]	31.6 [8.8]	.1	.50

Mean [Standard Deviation]. **Bold text** indicates means differed significantly at $p < .05$ in two-tailed, independent samples t-tests.

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Table 4.

Multivariable Models of Adolescent/Young Adult (AYA) Cancer Cognitions and Quality of Life Outcomes associated with Child and/or Maternal Factors

	B	SE *B*	R^2	*F*	*p*
Outcome | Determinants					
Perceived Cancer Risk			.10	10.13	<.001
AYA Age ≥18 years *	.24	.14			.09
AYA Female Gender **	.56	.15			<.001
Maternal Breast/Ovarian Cancer History ***	−.53	.17			.002
Cancer Knowledge (Total)			.04	5.46	.005
AYA Age ≥18 years *	.75	.33			.02
Maternal Breast/Ovarian Cancer History ***	−.83	.38			.03
Cancer Knowledge (Causes)			.07	9.79	<.001
AYA Age ≥18 years *	.67	.23			.004
Maternal Breast/Ovarian Cancer History ***	−.83	.27			.002

*<18 years = 1, ≥18 years = 2;

**Male = 1, Female = 2;

***1 = Yes, 2 = No